
Enterprise Value of firms in Insolvency

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Distress

A company faces uncertainty throughout its entire life cycle. However, uncertainty is high during both the high growth and declining stage of the company as the future is not fully predictable. For a firm under facing decline, uncertainty is magnified due to worsening of financial or economic conditions of the company. Stagnant or declining revenues, shrinking or negative margins, asset divestures, financial leverage, liquidity constraints are some of the factors that result a company to be distressed.

Financial Distress versus Economic Distress

Firms in financial distress cannot meet, or has difficulty paying off, its financial obligations to its creditors, typically due to high fixed costs, illiquid assets or revenues sensitive to economic downturns etc and it is reasonable likely that the company will become insolvent. It is directly related to the firm's leverage decision. On the other hand, firms facing economic distress are characterized by low or negative operating profitability and have questionable going concern value even in the absence of high leverage. When a firm is economically distressed, it means that the assets are not in their highest value use and shutting down the operations and divesture of assets / liquidation of assets is more beneficial for the company. Generally, insolvency or bankruptcy of a company results from financially distressed company.

It has generally been observed that companies in economic distress condition are unlikely to be revived as their products and processes are obsolete compared to the companies under financial distress.

Recognition of a distress company in India?

Many countries use different criteria to decide whether the company is distressed or not like Chinese listed companies defines financial distress when their profits continue to be negative for two consecutive years. For Iranian companies, distress is where retained losses are more than 50 per cent of their capital. Since no formal definition is provided in order to identify a company under distress in India, RBI in its circular¹ used a set of ratios which the banks can consider in order to model distress for its borrowers. Also, it mentions that a company is defined to be financially distressed if its accumulated losses are more than 50 percent² of its net worth³

Levels of Value

Levels of value depends upon the business ownership interest, and the purpose of the valuation. Hence, fair value of an asset or a company can be based primarily on two different levels:

- o *Marketable vs Non- Marketable:* Under this value, value represents how much investor can sell a company's shares and convert that investment in cash.
- o *Control vs Lack of control:* Under this value, value represents how much control does the investor / shareholder has over the company's business operations and strategic directions.

¹ RBI WPS (DEPR): 10/2016: Modelling Corporate Sector Distress in India

² According to Sick Industrial Companies Act, an industrial unit is regarded as potentially sick or weak unit if at the end of any financial year, it has accumulated losses equal to or exceeding 50 per cent of its average net worth in the immediately preceding four financial years and has failed to repay debts to its creditor(s) in three consecutive quarters on demand made in writing for such repayment. In the paper, similar definition of distress has been taken with some modifications.

³ Equity or Net worth comprises (a) paid-up capital, (b) forfeited shares (c) reserves and surplus, (d) money received against share warrants and (e) share application money pending allotment.

Enterprise Value v/s Liquidation Value

As a rule, a company is valued based on the premise that it would generate the greatest return to the owner of the property, taking into account what is physically possible, financially and legal feasible. Valuation of businesses in insolvency is important to the creditors, shareholders, and other stakeholders as a basis for negotiations between these parties over the future of the business. The two routes that a company under insolvency can take are

- a) to restructure its debts to manageable levels and continue trading as a going concern, or
- b) to liquidate the business, selling off business or physical assets piecemeal, and returning the proceeds to creditors

Enterprise Value

Enterprise Value ("EV") is a measure of a company's total value, represented by market capitalization plus debt, minority interest and preferred shares, minus total cash and cash equivalent. EV or going concern value is based on the premise that the firm will survive and return to financial health in the future. Hence, for a firm under insolvency, estimating an EV is essential to realize the greatest value to owners of the property. This can be achieved through potential sale to either a strategic buyer or a financial buyer or continuing as a going concern entity after certain corporate restructuring has been done.

A business which is merely financially distressed (i.e. one whose assets are more valuable if kept together as a functioning unit than they would be if sold off piecemeal) is said to have a 'going concern surplus'⁴. One way of preserving this surplus is to dispose of the business as a going concern and thus to obtain the going concern value (also called 'enterprise value'):

A widely-used valuation benchmark is the concept of Fair Market Value, defined by the American Society of Appraisers as⁵ "The price, expressed in terms of cash equivalents, at which property would change hands between a hypothetical willing and able buyer and a hypothetical willing and able seller, acting at arm's length in an open and unrestricted market, when neither is under compulsion to buy or sell and when both have reasonable knowledge of the relevant facts" However, fair value for a company under liquidation would be similar to the fair market value of the Company as the value for a company under insolvency would be based on the assumptions that the buyer(s) are reasonably informed about the company is under distress.

Liquidation Value

Liquidation Value has been defined under various acts as listed below:

As per section 35 (1) of the Insolvency and Bankruptcy Code, 2016 ("IBC"), "Liquidation Value is the estimated realizable value of the assets of the corporate debtor if the corporate debtor were to be liquidated on the insolvency commencement date". Further, section 35 (2) of IBC requires the valuer to determine liquidation value using internationally accepted valuation standards.

⁴ Value over and above the liquidation value of assets of the company

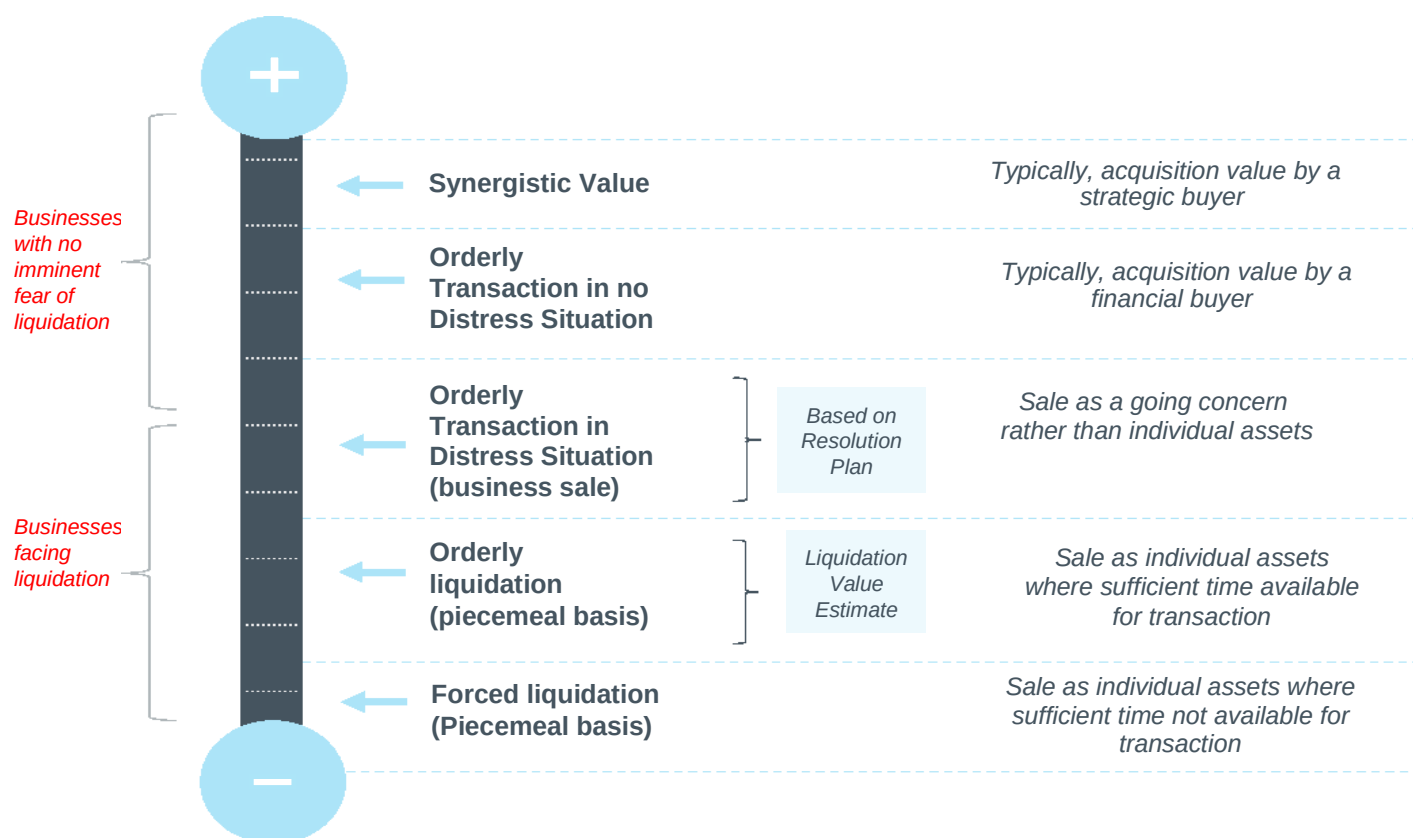
⁵ ASA Business Valuation Standards (Business Valuation Committee American Society of Appraisers, Herndon, VA, 2009), American Society of Appraisers p 27

According to the International Valuation Standards (“IVS”) 104, “Liquidation Value is the amount that would be realized when an asset or group of assets are sold on a piecemeal basis, that is without consideration of benefits (or detriments) associated with a going-concern business”.

According to the Indian Banks’ Association (IBA),” Liquidation Value describes the situation where a group of assets employed together in a business are offered for sale separately, usually following a closure of the business”.

Therefore, Liquidation value is essentially, the estimated value of assets when sold on piecemeal basis on the insolvency commencement date whereas the EV or the going concern value is the value estimated based on the assumption that the business will continue in the near future with some risk of distress / insolvency is assumed at the insolvency commencement date.

Below represents a quick overview of the levels of Value for a company under distress:



Estimation of Enterprise Value of a firm under distress & insolvency

Estimating the enterprise value or the going concern value of the distressed companies using traditional valuation methods, which are more suitable to healthier companies both economically and financially and where going concern assumption is not uncertain, must be suitable modified with assumptions reflecting risk of bankruptcy and uncertainty of operations. There are several ways to incorporate the risk of bankruptcy while estimating the enterprise value of the firm under different valuation approaches which includes –

1. Income approach - Modified Discounted Cash Flow method
2. Income approach – Going concern valuation and adjusting the distress separately
3. Relative approach based on the comparable companies and transactions

Below sections discusses below the approach required to estimate the value of firm under distress/ insolvency:

Income Approach – Modified Discounted Cash Flow Method

Discounted cash flow method is an intrinsic value and is theoretically represents the true value of an asset. For a company in distress, discounted cash flow method is feasible only when the restructuring plan makes detailed assumptions about the cash flows during the transition period from distress to financial health as the enterprise value of the firm under this method is clearly linked to those assumptions. Many restructuring plans are over optimistic and do not account for any possibility of distress thereby estimating the future cash flows. Hence, building the effects of the risk of distress into the discounted cash flow, probability of default i.e. the firm will not survive into the expected cash flows. Hence, a scenario based valuation method would be considered, ranging from the most optimistic (the company performs well and capital structure is optimal) to the most pessimistic (the company is surviving and making the lowest margins in the industry), assign probabilities to each scenario and estimate cash flows under each scenario, and then estimate the expected cash flows each year.

$$\text{Expected cash flow} = \sum_{j=1}^{j=n} \pi_{jt} (\text{Cashflow}_{jt})$$

where π_{jt} is the probability of scenario j in period t and Cash flow_{jt} is the cashflow under that scenario and in that period. These inputs have to be estimated each year, since the probabilities and the cash flows are likely to change from year to year. Note that the adjustment for distress is a cumulative one and will have a greater impact on the expected cash flows in the later year. Thus, if the probability of distress is 10% in year 1, the expected cash flows in all subsequent years have to reflect the fact that if the firm ceases to exist in year 1, there will be no cash flows later. If the probability of distress in year 2 is 10% again, there is now only an 81%⁶ chance that the firm will have cash flows in year 3. Under this method, the effects of distress are incorporated in both the expected cash flow and discount rates (discussed in discount rate section)

Simulation based approach of estimating the cash flows

Alternatively, distress can also be incorporated into the income approach using simulation analysis of the key variables of the business such as revenue growth, profitability etc. This will also take into account the possibility of negative outcomes or cause the firm into liquidation. Simulation can be performed using Monte Carlo simulation technique or probability simulation. It provides for a range of possible outcomes and the probabilities for each course of action including extreme possibilities (liquidation of the company) and for the most conservative decision (firm will continue as a going concern entity with reasonable profitability and optimal capital structure) - along with all possible consequences for middle-of-the-road decisions. Following steps need to be performed for the simulation process:

- o Choose the variables in the DCF valuation that you want estimate probability distributions on such as revenue, profitability, capital structure etc.
- o Define the distributions (type and parameters) for each of these variables.

⁶ Probability of surviving into year 3 = $(1-.10) (1-.10) = 0.81$

- o Run a simulation, where you draw one outcome from each distribution and compute the value of the firm. If the firm hits the “distress conditions”, value it as a distressed firm.
- o Repeat the previous step as many times as you can
- o Estimate the expected value across repeated simulations and then an average of the expected values are taken into consideration for the estimation of the enterprise value of the distressed firm.

Estimating the terminal value

The terminal value represents the value of the cash flows after the initial projection period into perpetuity. Estimating the terminal value is critical as it forms the largest part of the enterprise value as terminal value implicitly assumes that any financial distress is temporary and that the firm will not cease operations in the future. This neglects the risk of bankruptcy and the possibility that the firm might liquidate and cease to exist which will lead to overestimation of the value of the firm. Further, a scenario based approach can also be considered to account for the risk of bankruptcy / distress into the valuation. For this purpose, the terminal value can be estimated using a combination of liquidation approach (liquidation value of assets and liabilities), multiple approach (based on the comparable companies) and stable growth perpetuity model.

Valuing distress separately from the going concern value of the firm⁷

An alternate way of incorporating probability of distress while estimating the value of the distressed firm is accounting for the distress separately from the going concern value. This is because the traditional methods of valuation do not work well for companies in declining stage and also facing financial distress. Hence, there is a need of a new framework of model which addresses the issue of bankruptcy or distress risk into the value of the distressed firm. Hence, the below adapts the identified issues and deal specifically with the risk of default in a separate variable, tailored to the firm's characteristics. The model is based around the following equation:

Value = Value Going Concern * (1 – probability of distress) + Liquidation Value * probability of distress
Where probability of distress represents the cumulative probability of distress over the valuation period.

The main idea behind the model is that the firm's value is given by the average of the firm's value as a going concern, i.e. the firm will continue to exist and its liquidation value, i.e. the firm's defaults, weighted by the firm's probability of default. Both the going concern value and the liquidation value are adapted to the challenges faced by firms in decline and distress.

Probability of default or distress

Probability of default (PD) is the risk that the borrower will be unable or unwilling to repay its debt in full or on time. The risk of default is derived by analyzing the obligor's capacity to repay the debt in accordance with contractual terms. PD is generally associated with financial characteristics such as inadequate cash flow to service debt, declining revenues or operating margins, high leverage, declining or marginal liquidity, and the inability to successfully implement a business plan.

⁷ Sebastian Afflerbach, Master Thesis on Valuation of distressed firms, August 2014

Estimating probability of default or distress

The probability of default is carefully derived using a metric called Distance to Default which is based on the Black-Scholes-Merton Options Pricing Model

1. **Z-score:** Under this method, probability of distress is based on the observing the historical database of firms for reasons that led to bankruptcy and estimate the likelihood of future bankruptcy based on certain observable characteristics such as firm size, leverage and profitability. Hence, a statistical approach of estimating the Z-score was developed by Altman (1968), where he provided a set of 5 ratios which could be used to forecast which firms could possibly go bankrupt. Following is a set of ratios used:

$$Z = 0.012 (\text{Working capital/ Total Assets}) + 0.014 (\text{Retained Earnings/ Total Assets}) + 0.033 (\text{EBIT/ Total Assets}) + 0.006 (\text{Market value of equity/ Book value of total liabilities}) + 0.999 (\text{Sales/ Total Assets})$$

Different set of ratios can be used depending upon the industry and the firm characteristics. The above linear discriminant analysis does not provide a probability of default hence a probit or a multiple regression model should be used where the dependent variable takes the value of zero (firm survived) or one (firm went bankrupt) and the independent variable being the financial ratios for a company. If the relationship between both the variables is statistically and economically significant, then we have the basis of estimating the probability of distress.

2. **Bond Rating:** Under this method, probability of distress is based on the probability of default or distress estimated by credit rating agencies for different rated bonds. For instance, Altman estimated the cumulative probabilities of defaults for bonds for different ratings over a period of five and ten years. Higher the cumulative probabilities for the company, higher the changes of bankruptcy and lower the value of the firm. For instance, as per the Altman's estimation, bond with BBB rating for a period of 10 years, has a cumulative probability of 19.63 percent. This model only estimates the likelihood of default on the bond, but do not indicate whether the distressed firm goes out of business.
3. **Bond Price:** Unlike the conventional approach to valuing bonds discounts promised cash flows back at a cost of debt that incorporates a default spread to come up with a price. Alternatively, we could discount the expected cash flows on the bond, which would be lower than the promised cash flows because of the possibility of default, at the risk-free rate to price the bond. If we assume that a constant annual probability of default, we can write the bond price as follows for a bond with fixed coupon maturing in N years

$$\text{Bond Price} = \sum_{t=1}^{t=N} \frac{\text{Coupon} (1 - \pi_{\text{Distress}})^t}{(1 + \text{Riskfree Rate})^t} + \frac{\text{Face Value of Bond} (1 - \pi_{\text{Distress}})^N}{(1 + \text{Riskfree Rate})^N}$$

Where pd of distress is the annual probability and cumulative probability of distress can be then estimated for the desired period.

4. **Distance to Default Method:** Under this method, the probability of default is defined as the probability that the market value of the firm's assets will fall below the book value of firm's debt by the debt matures.

$$\text{Distance to Default} = \frac{\text{Market Value of Assets} - \text{Default Point}}{\text{Market Value of Assets} * \text{Asset Volatility}}$$

Lesser the distance to default, more the probability of default and vice versa. Default point is set equal to the book value of debt. Market value of assets is estimated based on the return on assets whereas the asset volatility is based on the firms' size, leverage, nature of the firm's business. This model can then be calibrated using the market value of the assets and the observed (or implied) equity price volatility in order to derive the Distance to Default or probability of default.

Discount Rate

One of the approach of incorporating the probability of insolvency / bankruptcy while using the income approach is adjust for this risk in estimation of the discount rate. Distressed firm will have higher costs of equity, higher costs of debt and usually have higher costs of capital than safer firms. If the distress is caused by high financial leverage, the cost of equity should be much higher.

We have explained below about the different considerations to be taken with respect to estimating the various components of a discount rate for companies under insolvency / distress:

Cost of Equity

In conventional valuation, we often estimate the cost of equity using a regression beta and the cost of debt by looking at the market interest rates on publicly traded bonds issued by the firm. For firms with a significant probability of distress, this approach can lead to inconsistent estimates. Consider first the use of regression betas. Since regression betas are based upon past prices over long periods (two to five years, for instance), and distress occurs over shorter periods, we will find that these betas will understate the true risk in the distressed firm. With the interest rates on corporate bonds, we run into a different problem. The yields to maturity on the corporate bonds of firms that are viewed as distressed reach extremely high levels, largely because the interest rates are computed based upon promised cash flows (coupons and face value) rather than expected cash flows. The presumption in a going concern valuation is that the promised cash flows have to be made for the firm to remain a going concern, and it is thus appropriate to base the cost of debt on promised rather than expected cash flows. For a firm with a significant likelihood of distress, this presumption is clearly unfounded.

Beta

Using a historically regressed beta for the distressed company can lead to underestimation of the cost of equity as regression betas lag distress as they are calculating over a long historical period hence, a using a bottom-up beta approach is suitable. This approach uses beta based on the peer group operating in the same industry adjusted for financial and operating risks between the distressed company and the peer set. These betas are then unlevered based on the financial and operating risks of the peer set and then relevered based on the financial and operating risks suitably. This method is useful for companies which are listed as well privately held.

Distress premium above normal publicly available equity risk premium

Distress premium represents the extra return demanded by potential buyers / investors of distressed assets depending upon the nature of the buyer. This is generally added separately to the cost of equity to arrive at the cost of equity for the company under distress / insolvency. This can be estimated using two ways:

- a) Looking at historical data on returns by investing in equity markets; or
- b) Compare distressed company's own current pre-tax cost of debt to the industry cost of debt.

Cost of Debt ⁸

For a distressed firm, the current market rate of borrowings is significantly higher due to current capital structure or bad economic conditions. For our purpose of estimating the cost of debt, the rate of borrowing should be taken into considered based on the future profitability and capital structure based on the restructuring plan. For this purpose, the cost of debt can be estimated based on the firm's bond rating, if the firm is rated otherwise, it can be estimated using the following ways:

- a) historical spreads on the borrowing of the distressed firm
- b) a synthetic rating can be estimated based on the financial characteristics of the firms within each class of ratings (for example, interest coverage ratio, capital structure etc)

Tax Rate

The current effective tax rate for a distressed company might be the minimum alternative. The tax rate is low in the beginning as the company as tax carry forwards might exist and increases as it starts generating profits and its situation improves. Considering that for a distress firm, the tax rate would gradually increase, it is suggested that effect of this is included while estimating post tax cost of debt by using different post tax cost of debt at different points of time during the projected period.

Capital Structure / Leverage

A distressed company is generally faced with a financial distress and hence, considering a constant target capital structure for the company for calculating the discount rate may not be appropriate as it the improvement of capital structure towards the industry average, although dependent on the restructuring plan, is mostly likely to be gradual over time. Hence, a separate discount rate needs to be estimated for each year till the industry average as per the restructuring plan is achieved. This can be done by varying the debt equity ratio and various other inputs used in the discount rate calculation for which the forecasts are available.

Company Specific Risk Premium:

A ratio analyses and common size financial statements should be performed to compare the subject company's current performance with its past performance as well as benchmarking to compare the subject company with industry and guideline company data. The financial analyst also considers nonfinancial risks such as those associated with the subject's industry, size (as measured by revenues and/or assets), depth of management (or lack thereof), operations, environmental laws and regulations, and so forth. The stage of distress, reorganization, and/ or restructuring is an additional consideration in determining the company specific risk premium.

Historical returns of ARC and distressed funds in India

Implementation of the 15/85 scheme (earlier 5/85) for the new capital requirements in August 2014 has resulted in upfront cash component increased to 15 percent, the ARCs would require more capital to acquire the stressed assets in India along with funding of working capital needs for the stressed assets. This has significantly increased the cost of acquisition of stressed assets for ARCs. This change also resulted in the ARC. This scheme has also decreased the dependence of the ARCs to rely just on the management fee based on the net asset value of the security receipts

⁸ Damodaran's analysis

issued to the investors (earlier estimated based on the outstanding security receipts). This has reduced the overall IRR for the ARC in India.

To offset this, ARCs have been seeking higher discounts to buy NPAs; however, banks are unwilling to reduce price, resulting in an expectation mismatch. This has led to a sharp decline in the transaction closure rate. The main reason for this gap appears to be the vastly different discounting rate used by banks and ARCs. While banks use discount rates in the range of 10% to 15%, given their access to cheap capital in the form of public deposits, ARCs use much higher discount rates of 20% to 25% as their cost of funds is relatively higher than that of banks. Without realistic valuation guidelines, there is no incentive for private investors to participate in auctions as the reserve price tends to be high, given the low discount rate used by banks vis-à-vis ARCs and private investors.

Factors which have affected might affect the return expected by the ARCs in India

NPA resolution in India is complex, tedious and time consuming. As per the World Bank's Doing Business 2018 report reveals that in terms of insolvency resolution, India holds 174th rank out of 212 countries. Further, it takes 4.3 years, on an average, for any resolution⁹. Also, India only recovers a mere 26% of its stressed assets due to following two reasons –

- o Slow judicial processes with repeated protracted appeals in higher courts of law
- o Lower market value of stressed assets than what the banks currently reflect on their balance sheets

Also, in February 2017, the Security Exchange Board of India's (SEBI's) nod to allow the listing of security receipts issued by ARCs on stock exchanges. This would bring in liquidity in the security receipts market and more money to ARCs, thereby resolving banks' NPA issues. Further, foreign capital flowing into the distressed market, ARCs may be willing to take more risks and bid for bigger assets on the hope that IBC will lead to faster resolution and recovery.

Reorganization costs

Reorganization costs are one-time costs incurred to extend the life of a company facing bankruptcy through special arrangements and restructuring in order to keep the business alive. This cost relates to the objective of getting the debt levels down and closer to their optimal capital structure. This can be achieved through either persuading the creditors to write down their claims, sell assets or issue new securities. The impact of reorganization costs in estimating the valuation of the distressed firm can be considered while estimating the discount rate where instead of using a single discount rate for all the years, different discount rates can be considered based on the varying capital structure for each period of cash flow. Higher discount rate in the first year would reflect higher leverage and then moving towards the optimal capital structure in the outer years thereby reducing the discount rate.

Relative Valuation Approach

Another method which can be adopted to estimate the enterprise value of the company under distress / insolvency is the relative valuation approach. Under relative valuation, the value of a distressed firm is based on market multiples for comparable firms, such as earnings, revenue or book value multiples or comparable transactions taken in the recent past. Under this method, there are two important steps while valuing the distressed firm:

- o Selection of comparables

⁹ World Bank's Doing Business 2018 Report

- o Selection of appropriate multiples and appropriate financial performance metric

Selecting the comparables

While valuing a company under distress, it is essential to select the right set of companies as there might be very few companies under similar situation operating in the same industry. Hence, the best option would be to include all the listed distressed firms in the comparable set irrespective of the industry and then adjusting the appropriate multiples based on the various factors like default risk, earnings, growth, asset base, debt equity ratio etc. This adjustment is done as the going concern assumption might be embedded in valuation of other listed comparable companies which are financial healthy than the subject distress company under valuation. Alternatively, incase no basis can be concluded while estimating our value estimate based on the comparable distressed firms, then valuation based on the multiples of the comparable healthy firms can be considered as a base and then separately incorporating a probability of default or distress for estimating the value of firm under distress along with liquidation value as the second scenario.

Distress-adjusted value = Relative value based on the healthy firms * (1- probability of distress) + liquidation value * (probability of distress)

Further, a comparable transactions method can be used to benchmark the value of the distress company under the other valuation methods. One may look at similar transactions of distressed assets in the past to estimate the value of company under distress / insolvency. However, factors like earnings, growth, asset base, debt equity ratio etc. needs to be considered while selecting the transaction as a comparable to estimate the value of the firm under distress.

Selecting appropriate multiples and appropriate financial performance metric

For a distressed firm, price to earnings and price to book value based multiples may not be an appropriate as they are negative for a distressed firm hence, considering revenues or earnings before interest, tax, depreciation and amortization (EBITDA) based multiple might be appropriate. As discussed above, there could be two approaches of valuing a distressed firm using comparable company's method, where one set of companies considered can be distressed (best option) and other set where the companies are financially healthy. Hence, it becomes important to assign an appropriate discount on the multiple of the comparable companies based on the several financial factors to be used for the subject distressed company under valuation as multiples of the comparable distressed companies are not subject to market forces of demand and supply hence the multiple based valuation under this scenario becomes more complex and less precise. On the other hand, an appropriate financial performance metric should be the projected revenue or EBITDA (based on the TEV study) of the subject company as our going concern assumption that the company will return to industry average is incorporated while estimating the distress adjusted enterprise value of the firm.

To arrive at the concluded enterprise value or going concern value of the distressed firm, value under the income and market approach will be weighted based on the reliability and the nature of the information available.